NACL (NETWORK ACCESS CONTROL LIST)

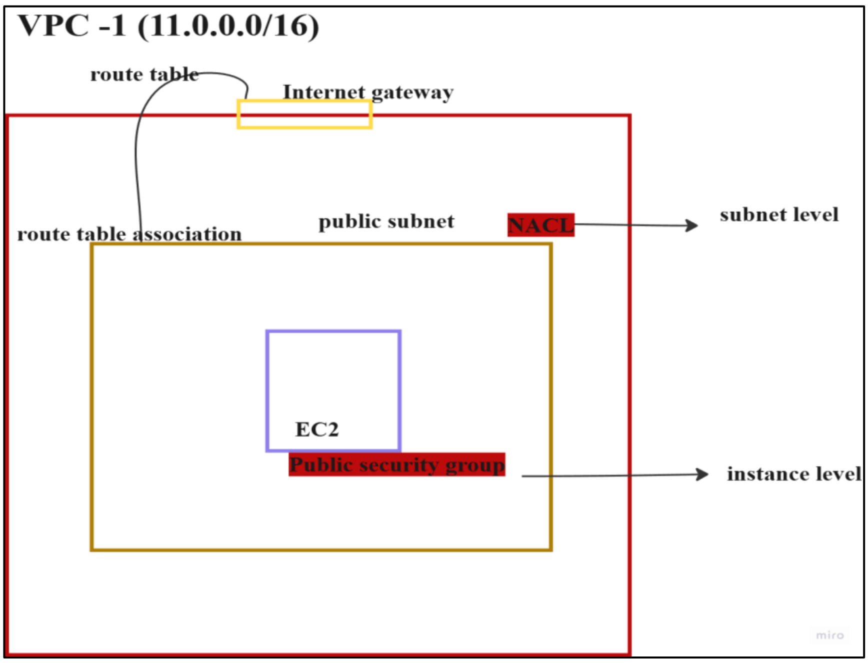
A Network Access Control List (NACL) is a stateless virtual firewall that operates at the subnet level in a VPC, controlling inbound and outbound traffic by using a set of ordered allow and deny rules.

SECURITY GROUP

A Security Group in Amazon Web Services is a stateful virtual firewall that controls inbound and outbound traffic to resources such as EC2 instances by allowing only specified traffic based on defined rules.

SECURITY GROUP VS NACL

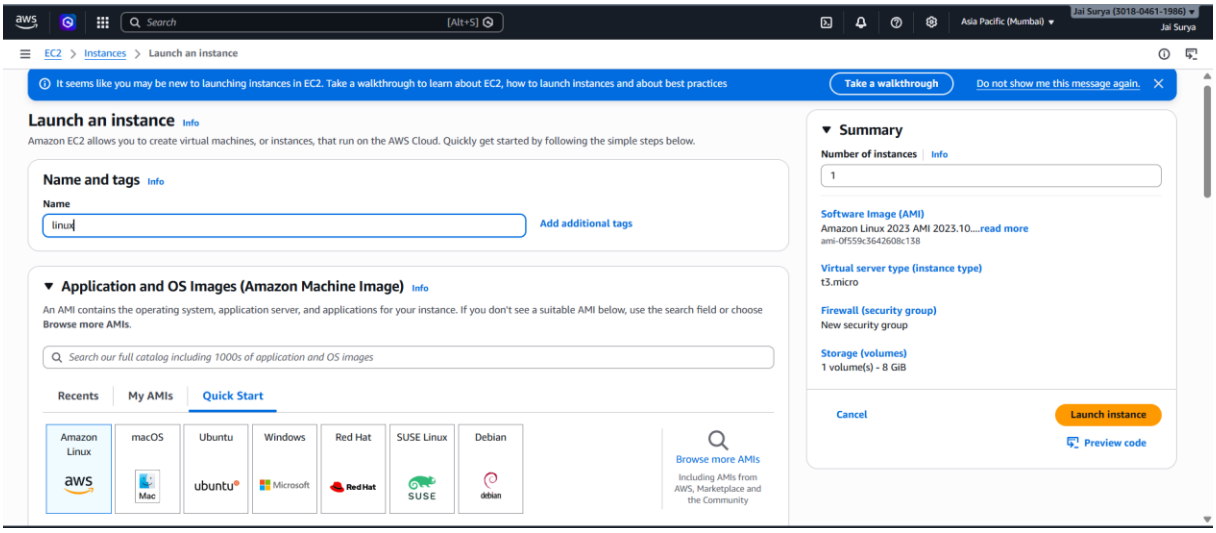
Feature	**Security Group**	**NACL (Network ACL)**
Level	Works at instance level	Works at subnet level
Rules	Allow only rules	Allow & Deny rules
State	Stateful (remembers traffic)	Stateless (no memory)
Traffic Check	Checks mainly inbound (auto allows return)	Checks inbound & outbound separately
Rule Processing	All rules evaluated together	Rules processed in number order

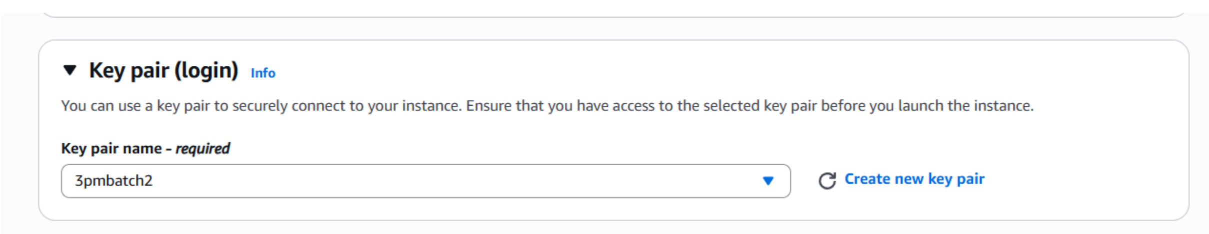


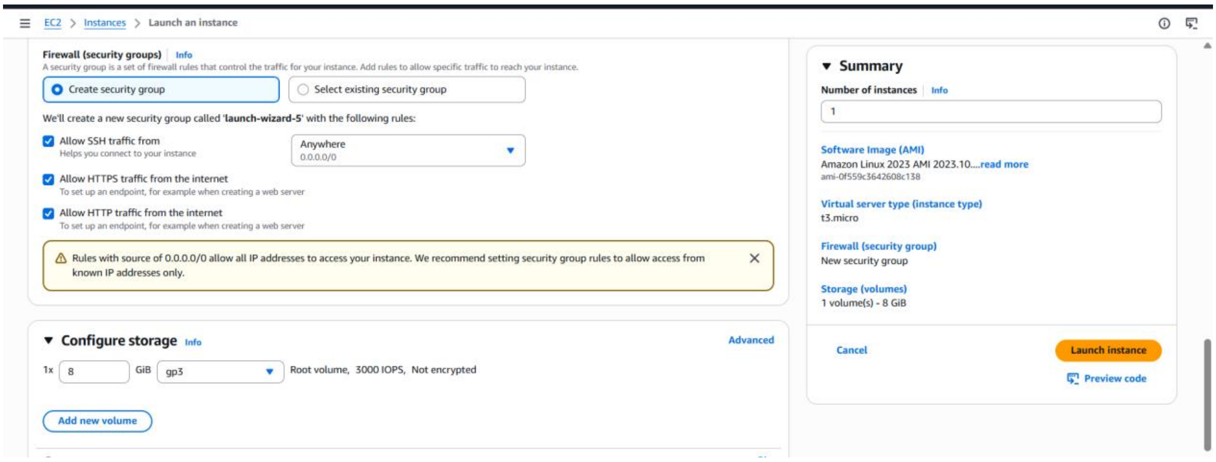
HANDS-ON FOR NETWORK ACL

Screenshot 1: Launch EC2 Instance

- Go to Amazon Web Services Console
- Navigate to EC2 Dashboard
- Click Launch Instance
- Enter instance name
- Select Amazon Linux AMI
- Choose/Create Key Pair
- Under Network settings, allow:
    - SSH (port 22)
    - HTTP (port 80)
    - HTTPS (port 443)
- Click Launch Instance

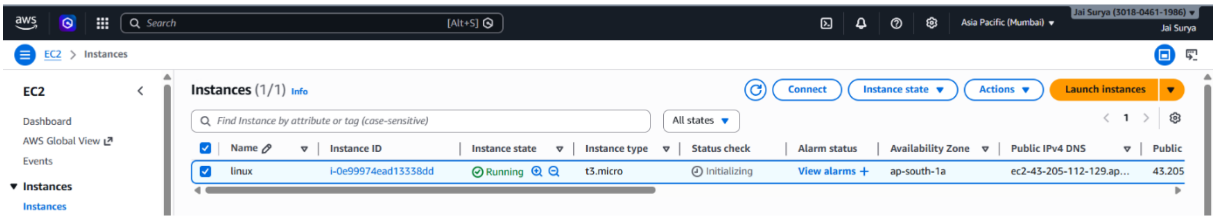






Screenshot 2: Connect to Linux Instance

- Go to **Instances**
- Select your Linux instance
- Click **Connect**

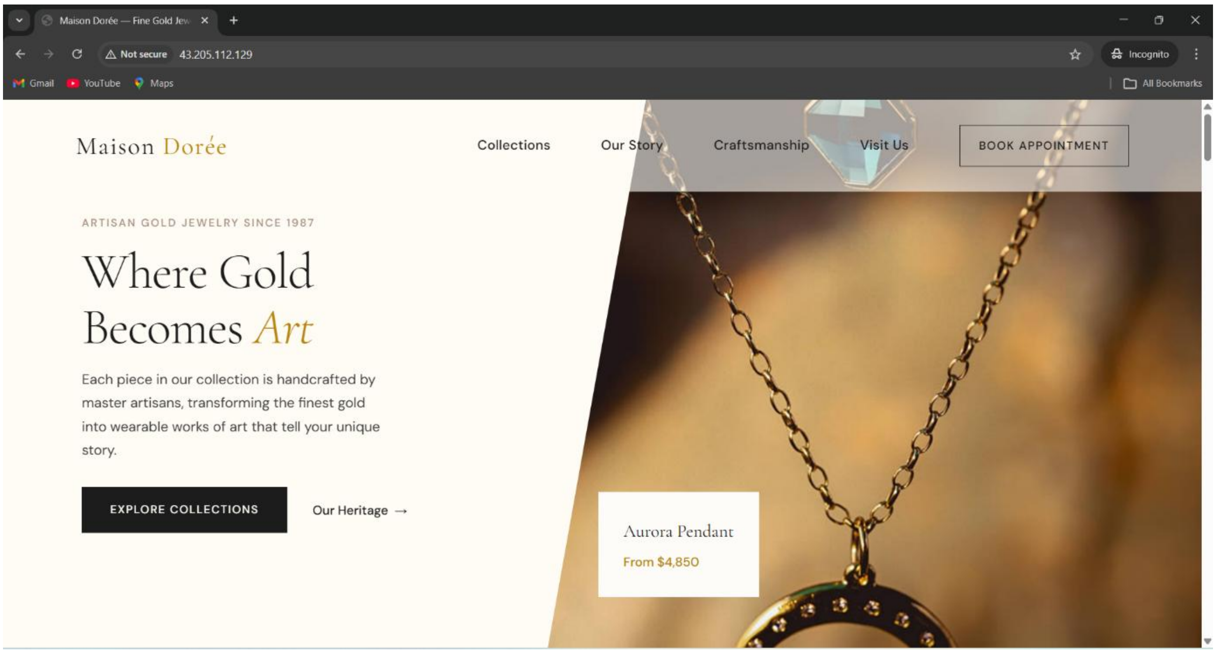


Screenshot 3: Execute HTTPD Web Hosting Commands

- Execute HTTPD web hosting commands in Linux terminal

Screenshot 4: Website Opened

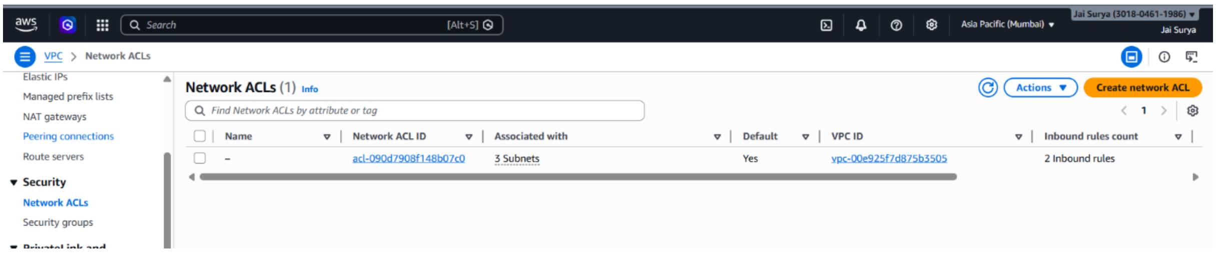
- Open the web browser
- Enter the public IP of the instance
- Verify that the website is displayed successfully



DENY RULE IN NACL

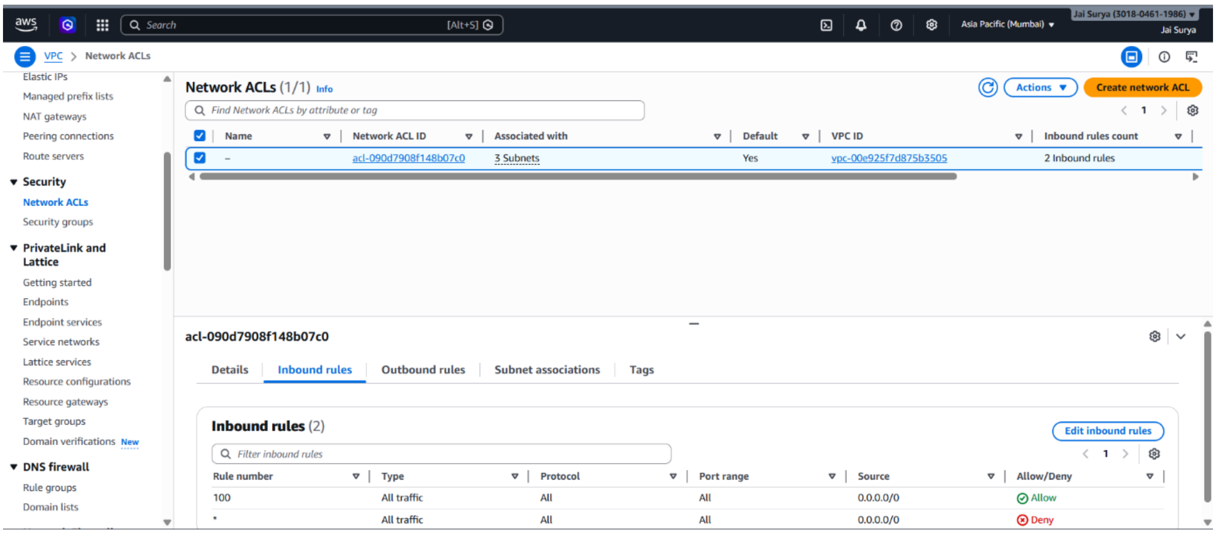
Screenshot 5: Network ACLs

- Go to **VPC**
- Click **Network ACLs**



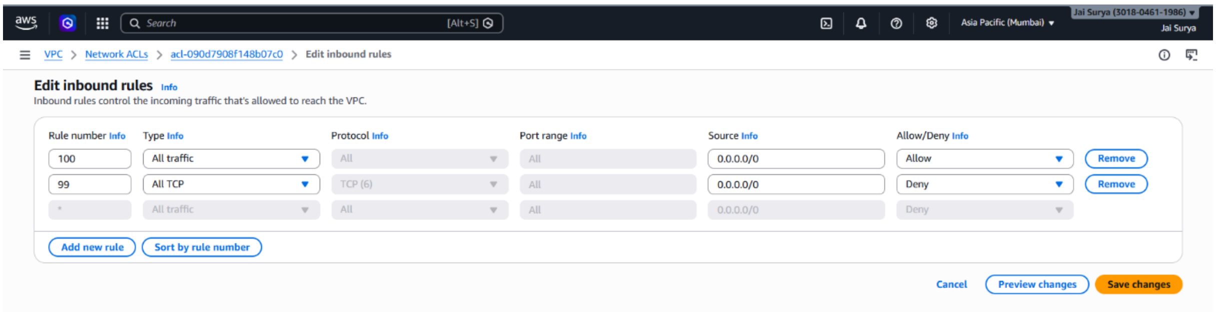
Screenshot 6: Edit NACL Inbound Rules

- Select **Default NACL**
- Go to **Inbound Rules**
- Click **Edit**



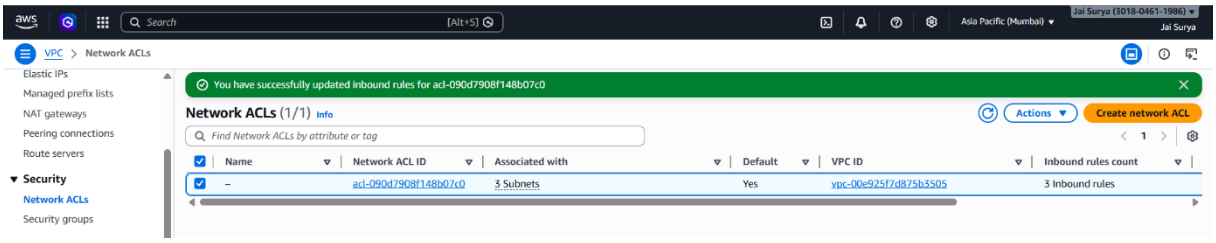
Screenshot 7: Add Deny Rule

- Click **Add Rule**
- Enter rule number **99**
- Select **All TCP**
- Set action to **Deny**
- Click **Save**



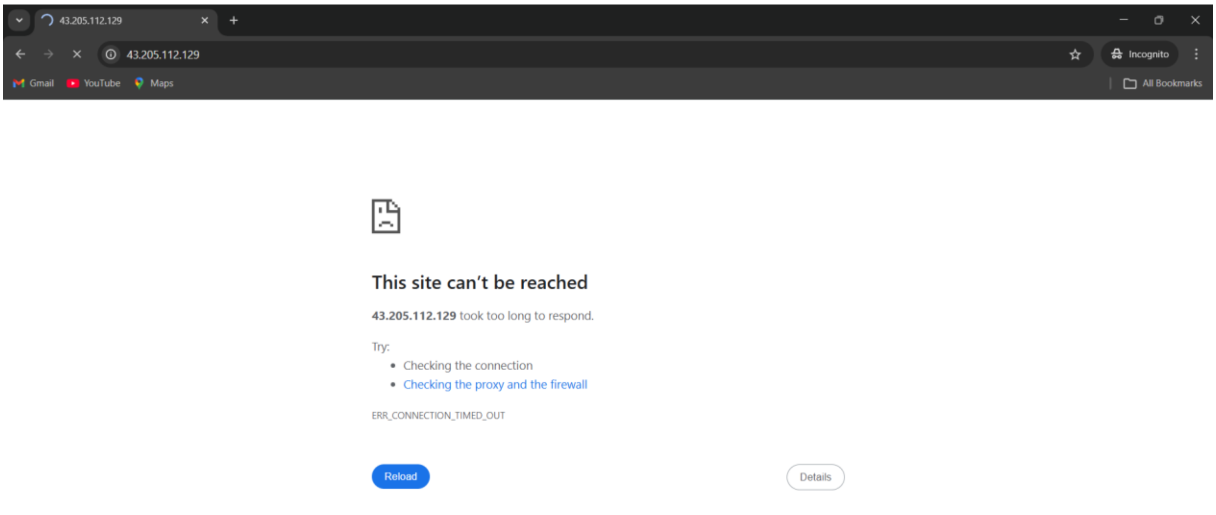
Screenshot 8: NACL Rule Updated Successfully

- Verify that the new rule is added
- Confirm the rule is updated successfully



Screenshot 9: Website Not Reachable

- Go to the web browser
- Refresh the website page
- Verify that the site is not reachable due to NACL deny rule



ELASTIC IP

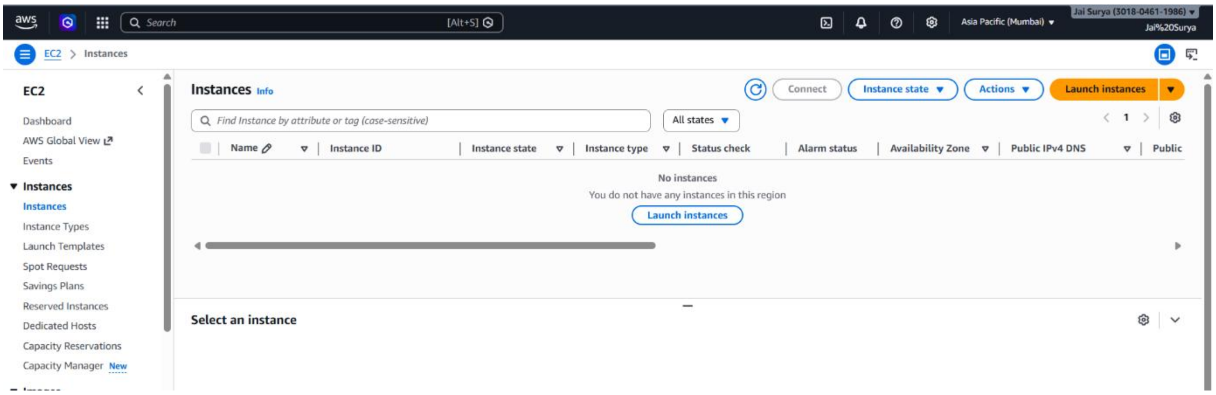
An Elastic IP in **Amazon Web Services** is a **static, public IPv4 address** designed for dynamic cloud computing, which can be associated with an EC2 instance and remains constant even if the instance is stopped or restarted.

HANDS-ON FOR ELASTIC IP

BEFORE USING ELASTIC IP

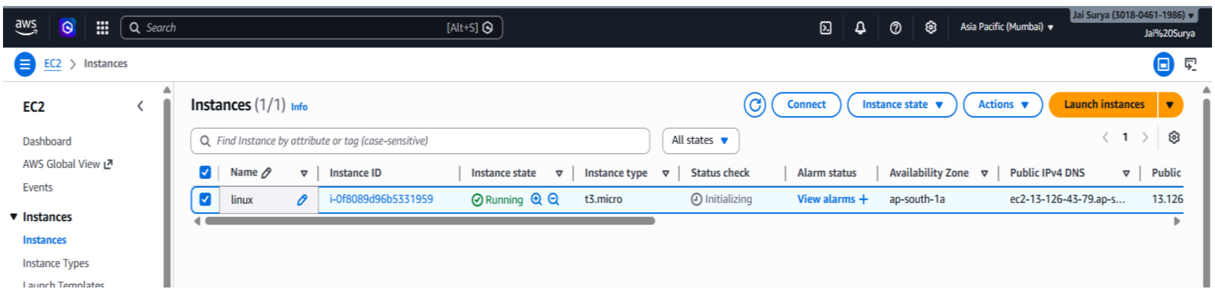
Screenshot 1: Create Linux Instance

- Go to Amazon Web Services Console
- Navigate to EC2
- Click Launch Instance
- Enter instance name
- Select Amazon Linux AMI
- Select key pair
- Click Launch Instance



Screenshot 2: Instance Created

- Go to Instances
- Verify that the instance is created and running

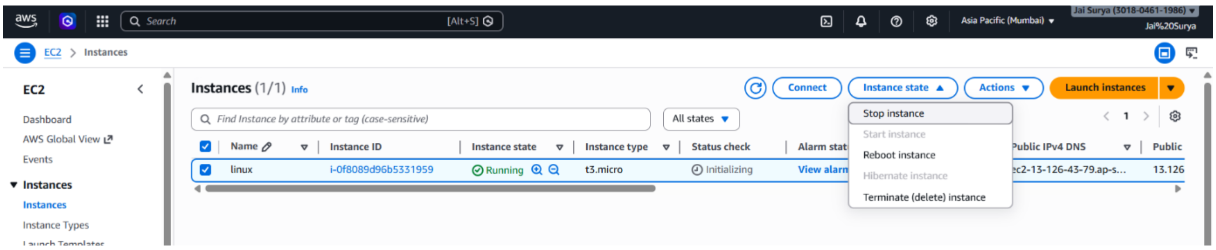


Screenshot 3: Verify Public IPv4 Address

- Go to Instances
- Select your instance
- Check the Public IPv4 address
- Verify the IP is assigned

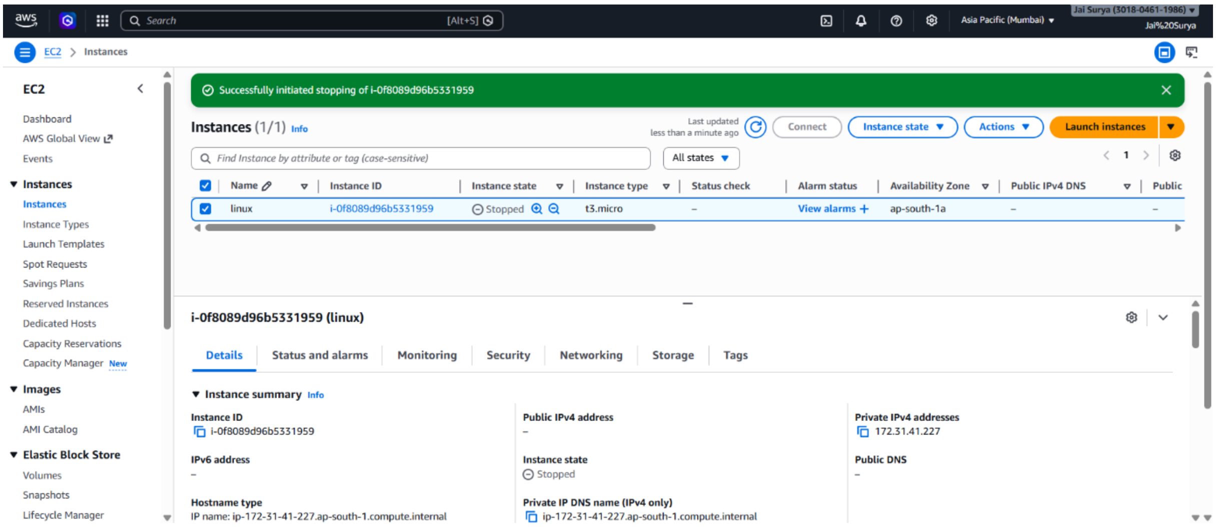
Screenshot 4: Stop Instance

- Select the instance
- Click Actions
- Click Instance State → Stop
- Confirm to stop the instance



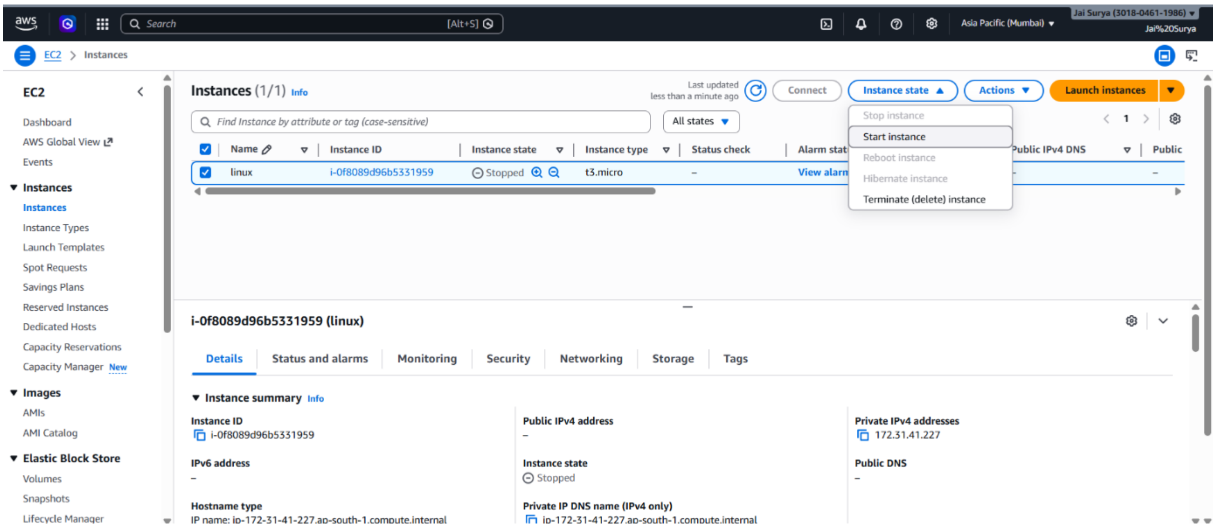
Screenshot 5: Public IPv4 Removed

- Verify that the instance is stopped
- Check the Public IPv4 address
- Observe that the public IP is removed



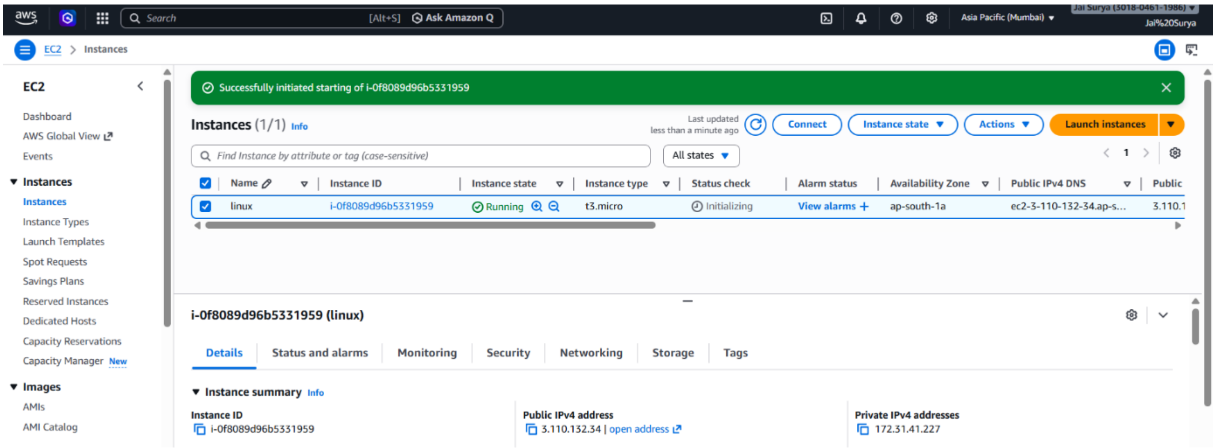
Screenshot 6: Start Instance

- Select the instance
- Click Instance State → Start
- Wait for the instance to run



Screenshot 7: Public IP Changed

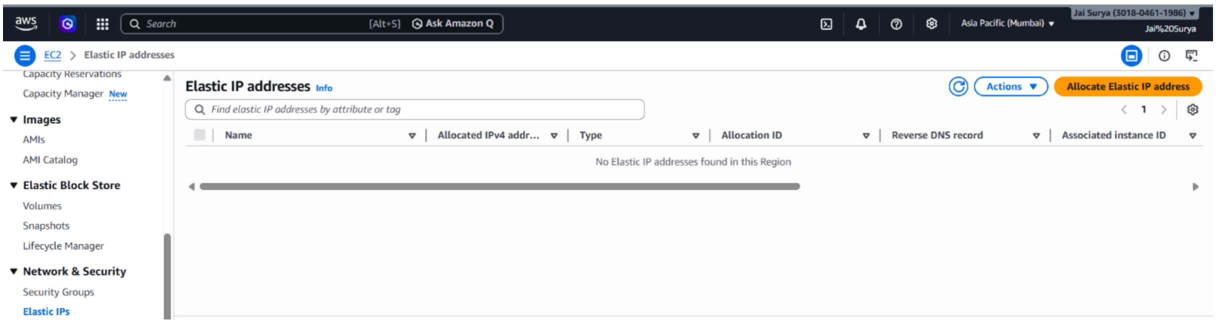
- Verify that the instance is running
- Check the Public IPv4 address
- Observe that the public IP has changed



AFTER USING ELASTIC IP

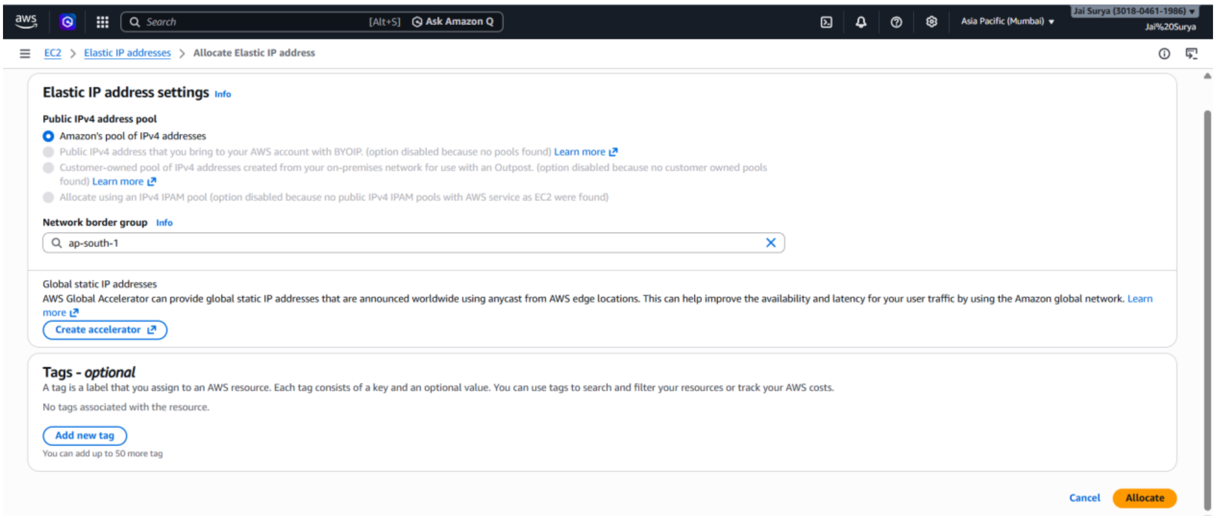
Screenshot 8: Allocate Elastic IP

- Go to Elastic IPs
- Click Allocate Elastic IP address



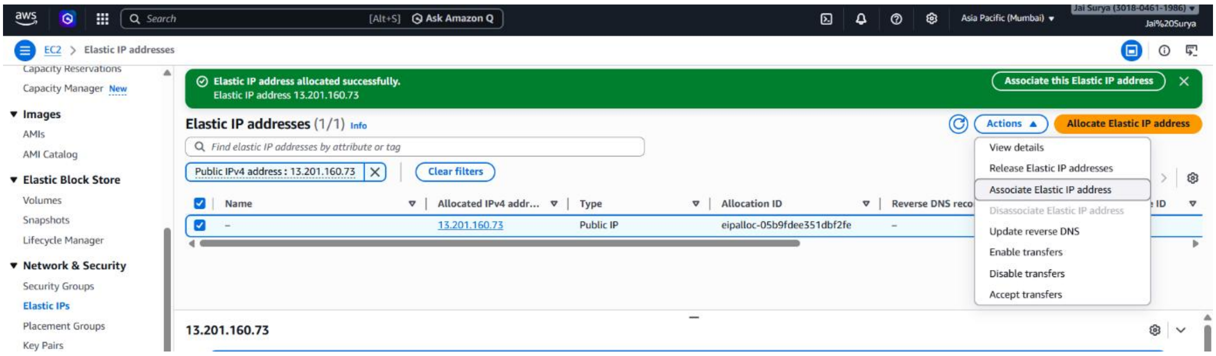
Screenshot 9: Select Network Border Group

- In Elastic IP allocation page
- Select Network Border Group
- Choose the required region
- Click Allocate



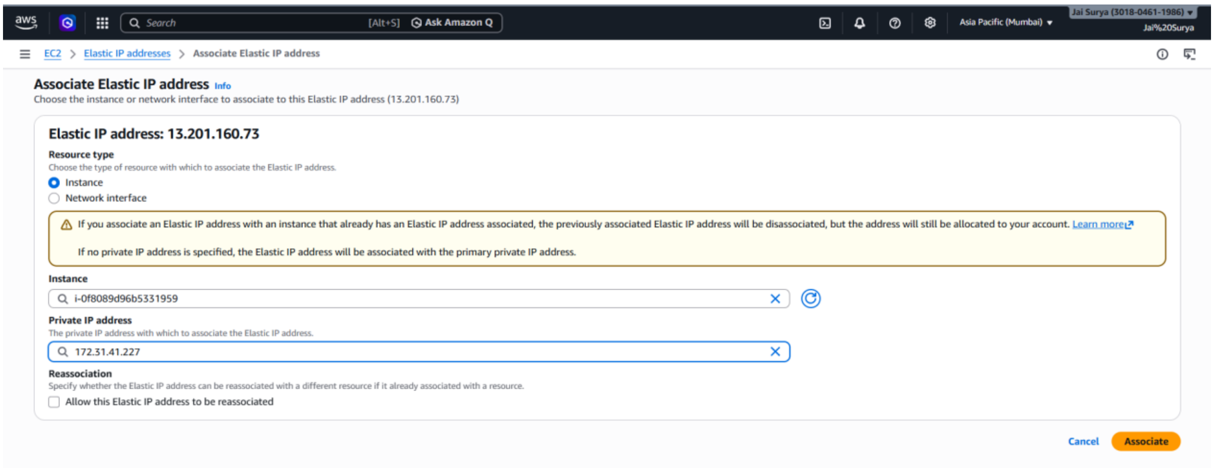
Screenshot 10: Associate Elastic IP

- Select the created Elastic IP
- Click Actions
- Click Associate Elastic IP



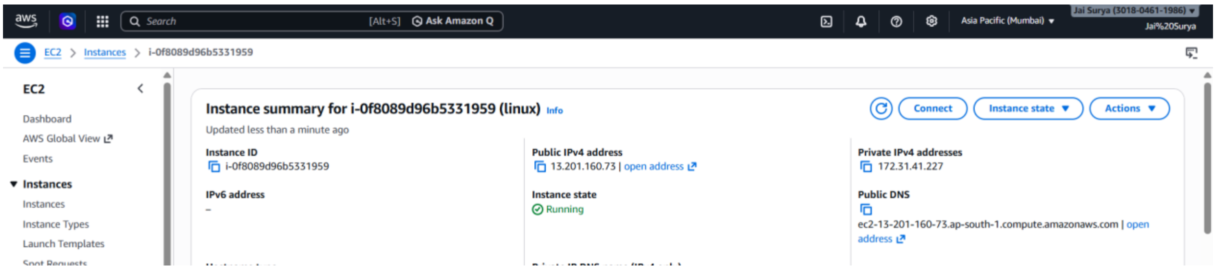
Screenshot 11: Select Instance and Private IP

- Select your instance
- Select the private IP address
- Click Associate / Enter



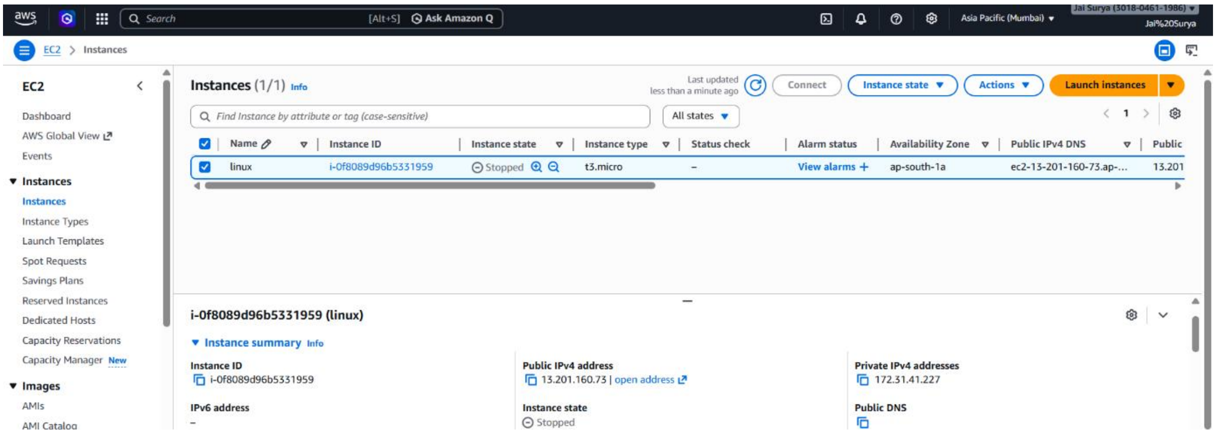
Screenshot 12: Verify Public IPv4 Address

- Go to Instances
- Select your instance
- Check the Public IPv4 address
- Verify that it matches the Elastic IP



Screenshot 13: Stop Instance and Verify IP

- Select the instance
- Click Instance State → Stop
- After stopping, check the Public IPv4 address
- Verify that the Elastic IP is not removed



Screenshot 14: Start Instance and Verify IP

- Select the instance
- Click Instance State → Start
- After starting, check the Public IPv4 address
- Verify that the Elastic IP remains unchanged